Mathematics for Machine Learning

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Artificial Intelligence and Machine Learning

Artificial Intelligence (AI)

Artificial Intelligence is a branch of computer science that focuses on inducing intelligence into machines and systems. By leveraging data and algorithms, AI enables systems to reason, learn, make decisions, optimize processes, and even generate creative content.

Machine Learning(ML)

Machine Learning (ML) is a branch of Artificial Intelligence (AI) that focuses on developing algorithms and models that allow computers to learn patterns and relationships from data and improve their performance on tasks without being explicitly programmed

When to USE Machine Learning ?

Properties of ML Problems

1. Problems with Uncertinity
2. Data with hidden patterns
3. Approximating for Generalization
4. Minimum error on unseen data

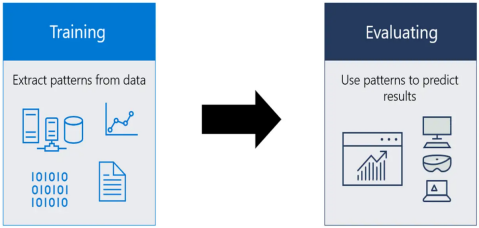
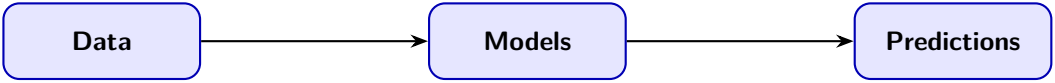


Figure: Machine Learning

Machine Learning Pipeline

ML Training

ML Inference



Why mathematics for Machine Learning

Key Idea

Data are points in $\mathbb{R}^n$, models are transformations in $\mathbb{R}^n$, and learning optimizes objectives defined via inner products, norms, and decompositions.

- **Representation:** vectors, matrices, tensors (images, text, graphs).
- **Transformation:** linear layers, attention, convolutions (as linear operators).
- **Optimization:** gradients, Hessians, curvature (matrix calculus).
- **Beyond Euclid:** kernels and manifolds still leverage $\mathbb{R}^n$ intuition.

Sets

Definition: A set is a **well-defined unordered** collection of **distinct** objects.

Examples:

- Set of natural numbers: $\mathbb{N} = \{1, 2, 3, \dots\}$
- Set of real numbers: $\mathbb{R}$
- Set of features in a dataset

Notation:

- $x \in A$: x belongs to set A
- $x \notin A$: x does not belong to set A

Sets Overview

Types of Sets

- Finite and Infinite Set
- Empty Set ($\emptyset$)
- Subset ($A \subseteq B$)
- Universal Set

Set Operations

- Union: $A \cup B$
- Intersection: $A \cap B$
- Difference: $A - B$
- Complement: A^c

Sets in ML

- Dataset = Set of samples
- Features = Set of variables
- Labels = Set of classes
- Train/Test $\subset$ Dataset

Algebraic Structures

Definition

An algebraic structure is a set equipped with one or more operations.

$$\text{Algebraic Structure} = \text{Set} + \text{Operation(s)}$$

Properties of Operations

- **Closure:** $\forall a, b \in G,\ a * b \in G$
- **Associativity:** $(a * b) * c = a * (b * c)$
- **Commutativity:** $a * b = b * a$
- **Identity Element:**
  $\exists e \in G$ such that $a * e = e * a = a$
- **Inverse Element:**
  $\forall a \in G,\ \exists a^{-1}$ such that $a * a^{-1} = e$

Examples

$$(\mathbb{Z}, +), \quad (\mathbb{R}, +, \cdot), \quad (\mathbb{R}^n, +, \cdot)$$

Types of Algebraic Structures

Group

- Closure and Associativity
- Identity
- Inverse

Example: $(\mathbb{Z}, +)$

Field

- Addition Multiplication
- Identity exists
- Inverses exist

Example: $\mathbb{R}, \mathbb{Q}$

Ring

- Addition Group
- Multiplication Closure

Example: $(\mathbb{Z}, +, \cdot)$

Vector Space

- Vector Addition
- Scalar Multiplication

Example: $\mathbb{R}^n$

Hierarchy of Algebraic Structures

Group $\subset$ Ring $\subset$ Field $\subset$ Vector Space

Vectors

Definition

In this context, a **vector** is an element of a *vector space*. It can be represented as an ordered collection of numbers (components), for example:

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \quad \text{or} \quad \vec{v} = (v_1, v_2, v_3).$$

Geometrically, vectors are arrows with *magnitude* and *direction*. They are not just numerical lists but fundamental objects inhabiting a mathematical space.

Mathematical Spaces

Definition

A **mathematical space** is a set of objects together with certain rules or structures that allow us to study their relationships. For example, a metric space defines how to measure distances.

Examples

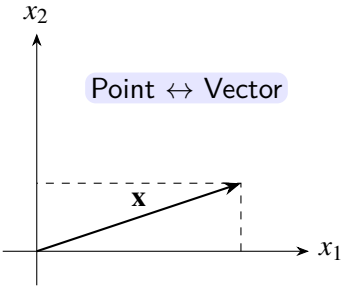
- **Metric Space:** The set of real numbers $\mathbb{R}$ with the distance $d(x,y) = |x-y|$.
- **Vector Space:** $\mathbb{R}^2$ with operations of vector addition and scalar multiplication.
- **Inner Product Space:** $\mathbb{R}^n$ with inner product $\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^{n} x_i y_i$.
- **Hilbert Space:** An infinite-dimensional space of square-integrable functions, L^2.

Vectors and Vector Spaces

Definition (Vector Space)

A set V over a field $\mathbb{F}$ with addition and scalar multiplication satisfying closure, associativity, commutativity, identity, inverses, and distributivity.

- Basis, span, dimension; subspaces and projections.
- Embeddings in NLP/CV: words/images $\mapsto$ vectors in $\mathbb{R}^n$.



Definition of a Vector Space

Axioms

These operations satisfy the following axioms for all $u, v, w \in V$ and $a, b \in \mathbb{F}$:

- $u + v = v + u$ (Commutativity)
- $(u + v) + w = u + (v + w)$ (Associativity)
- $\exists 0 \in V$ s.t. $v + 0 = v$ (Additive identity)
- $\exists -v \in V$ s.t. $v + (-v) = 0$ (Additive inverse)
- $a(u + v) = au + av$ (Distributivity)
- $(a + b)v = av + bv$ (Distributivity)
- $a(bv) = (ab)v$ (Associativity of scalar mult.)
- $1 \cdot v = v$ (Identity element)

$\mathbb{R}^n$ as a Vector Space

$(\mathbb{R}^n, +, \bullet)$

The set

$$\mathbb{R}^n = \{(x_1, x_2, \ldots, x_n) \mid x_i \in \mathbb{R},\ 1 \leq i \leq n\}$$

is a vector space over $\mathbb{R}$ with:

$$(x_1, \ldots, x_n) + (y_1, \ldots, y_n) = (x_1 + y_1, \ldots, x_n + y_n),$$

$$c \cdot (x_1, \ldots, x_n) = (cx_1, \ldots, cx_n), \quad c \in \mathbb{R}.$$

Geometric view: Each vector in $\mathbb{R}^n$ is an ordered n-tuple. For $n = 2$, vectors are arrows in the plane; for $n = 3$, they are arrows in space; for higher n, they are abstract but follow the same rules.

Inner Product and Norm

Inner Product

An inner product is a function that takes two vectors and returns a scalar.

$$\langle x, y \rangle = \sum_{i=1}^{n} x_i y_i$$

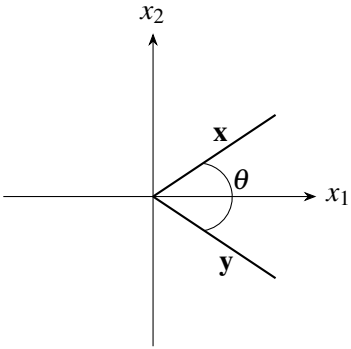
Properties:

- Symmetry: $\langle x, y \rangle = \langle y, x \rangle$
- Linearity: $\langle ax + by, z \rangle = a\langle x, z \rangle + b\langle y, z \rangle$
- Positive definite: $\langle x, x \rangle \geq 0$

Norm

Norm represents the length (magnitude) of a vector $\|x\| = \sqrt{\langle x, x \rangle} = \sqrt{\sum_{i=1}^{n} x_i^2}$

Inner Products and Norms



$$\cos\theta = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\| \, \|\mathbf{y}\|}$$

Inner Product Space

Definition

An **inner product space** is a vector space V over $\mathbb{F}$ (where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$) equipped with a function

$$\langle \cdot, \cdot \rangle : V \times V \to \mathbb{F},$$

called the *inner product*, such that for all $u, v, w \in V$ and scalars $a, b \in \mathbb{F}$:

- $\langle v, v \rangle \geq 0$ and $\langle v, v \rangle = 0 \iff v = 0$ (Positivity)
- $\langle au + bw, v \rangle = a\langle u, v \rangle + b\langle w, v \rangle$ (Linearity)
- $\langle u, v \rangle = \overline{\langle v, u \rangle}$ (Conjugate Symmetry)

Induced concepts:

$$\|v\| = \sqrt{\langle v, v \rangle}, \quad u \perp v \iff \langle u, v \rangle = 0$$

Example of an Inner Product Space

$\mathbb{R}^n$ as an Inner product Space

For vectors $u = (u_1, u_2, \ldots, u_n)$, $v = (v_1, v_2, \ldots, v_n) \in \mathbb{R}^n$, define

$$\langle u, v \rangle = \sum_{i=1}^{n} u_i v_i$$

(the usual dot product).

Special case in $\mathbb{R}^2$:

$$\langle (x_1, y_1), (x_2, y_2) \rangle = x_1 x_2 + y_1 y_2$$

Geometric meaning:

$$\langle u, v \rangle = \|u\| \, \|v\| \cos \theta$$

where θ is the angle between u and v.

Norm of a Vector

Definition: Norm

In an inner product space $(V, \langle \cdot, \cdot \rangle)$, the **norm** of a vector $v \in V$ is defined as

$$\|v\| = \sqrt{\langle v, v \rangle}.$$

Properties of norm:

- $\|v\| \geq 0$, and $\|v\| = 0 \iff v = 0$ (Non-negativity)
- $\|\alpha v\| = |\alpha| \, \|v\|$ (Homogeneity)
- $\|u + v\| \leq \|u\| + \|v\|$ (Triangle inequality)

Example in $\mathbb{R}^n$:

$$\|(x_1, x_2, \ldots, x_n)\| = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}.$$

Orthogonality

Definition: Orthogonality

Two vectors $u, v \in V$ are said to be **orthogonal** if

$$\langle u, v \rangle = 0.$$

If, in addition, $\|u\| = \|v\| = 1$, they are called **orthonormal**.

Example in $\mathbb{R}^2$:

$$u = (1,0), \quad v = (0,1) \quad \Rightarrow \quad \langle u, v \rangle = 0.$$

Geometric meaning: Orthogonal vectors are *perpendicular* to each other.

What is a Matrix?

Definition

A **matrix** is a rectangular array of numbers that can represent a linear transformation:

$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

Linear Transformations

Definition

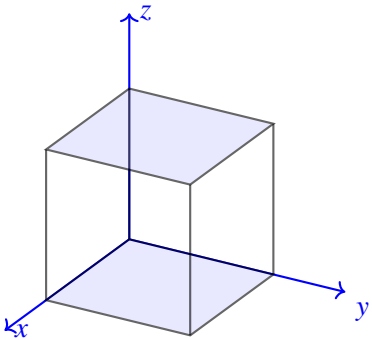
A **linear transformation** T preserves addition and scalar multiplication:

$$T(\vec{a}+\vec{b}) = T(\vec{a}) + T(\vec{b}), \quad T(k\vec{a}) = kT(\vec{a})$$

Examples in 3D

Scaling, *Rotation*, and *Shear*.

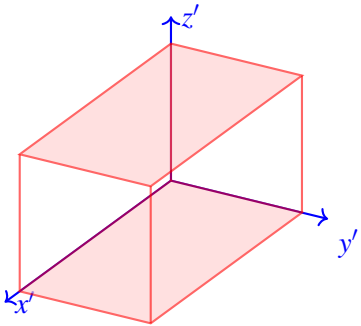
3D Example: The Unit Cube



Unit Cube in $\mathbb{R}^3$:

- Spanned by basis vectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$.
- Each edge has length 1.
- Volume $= 1$ (determinant of identity matrix).
- Serves as the reference object to visualize transformations.

Transformation Example: Scaling

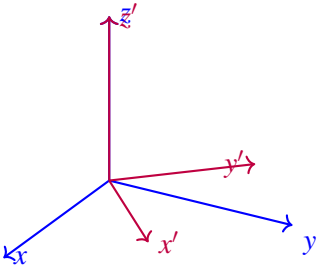


$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Effect:

- Stretches space by factor 2 along the x-axis.
- y, z axes remain unchanged.
- Volume scaling $= \det(A) = 2$.
- The unit cube becomes a cuboid of volume 2.

Transformation Example: Rotation



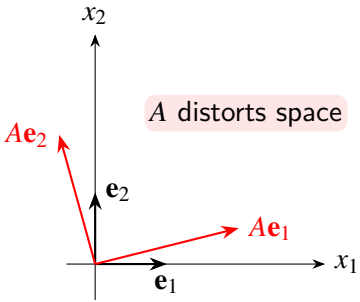
$$R_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \theta = 45^\circ$$

Effect:

- Rotates vectors in the xy-plane by 45°.
- z-axis remains fixed.
- Volume preserved: $\det(R_z) = 1$.
- Orientation preserved (no reflection).

Matrices and Linear Transformations

Definition

A matrix $A \in \mathbb{R}^{m \times n}$ represents a linear map $T_A : \mathbb{R}^n \to \mathbb{R}^m$, $T_A(\mathbf{x}) = A\mathbf{x}$.

- Composition = matrix multiplication; identity, orthogonal, diagonal matrices.
- Neural layers: $\mathbf{h} = W\mathbf{x} + b$, attention $\propto QK^\top$.